

ENGR. CELPHY TRIMUCHA,ECT

Licensed Electronics Engineer (RA 1080) |
MSECE (Candidate) Digital Laboratory Manager |
STI Innovation Champion

📍 Upper Barras, Bantayan, Kabankalan City, Negros Occidental

☎ 0945-384-3695 ✉ trimuchacelphy06@gmail.com

PROFESSIONAL PROFILE

Licensed Electronics Engineer (RA 1080) and research-driven innovator specializing in embedded systems, microcontroller applications, and sustainable engineering solutions. Experienced in Scopus-indexed research presentations, prototype development, laboratory research management, and mentoring student researchers. Actively engaged in applied research projects involving environmental technology, automation systems, and smart devices.

EDUCATION & ELIGIBILITY

- **Master of Science in Electronics Engineering (MSECE)** | University of St. La Salle (Ongoing)
- **Bachelor of Science in Electronics and Communications Engineering** | Fellowship Baptist College (2017)
- **Professional Eligibility: Licensed Electronics Engineer (RA 1080)** – Equivalent to Career Service Professional / Second Level Eligibility.
- **Safety Officer II. Basic Occupational Safety and Health Certificate Holder**

CORE COMPETENCIES

- **Functional:** Property & Inventory Accountability, Technical Procurement Management, Asset Lifecycle Management, Tech-Transfer & Commercialization.
- **Leadership:** STI Advocacy, Leading and Championing S&T Innovation, Strategic Planning, and Coaching for Results.
- **Core:** Advanced Problem Solving, Technical Communication (Scopus-level), and Use of Technology.

PROFESSIONAL EXPERIENCE

DIGITAL LABORATORY IN-CHARGE / COLLEGE INSTRUCTOR

2019 – Present Fellowship Baptist College – Kabankalan City

- **Property Management :** Acts as the primary custodian for all digital laboratory assets; implements strict inventory control, periodic auditing, and maintenance of high-value electronic equipment.
- **Procurement Management:** Leads the technical specification development for the acquisition of engineering components and STEM laboratory supplies, ensuring compliance with institutional standards.
- **STI Championing:** Mentored student teams to victory in the 3rd Kabankalan ICT Summit (FishDryPro - 2nd Runner-up & Most Voted Start-up).
- **Innovation & Research:** Developed and presented Scopus-indexed research (EcoPET) at the 7th ICAMEROB 2025, securing 2nd Best Presenter and 3rd Best Research Paper.

KEY RESEARCH & INNOVATION PROJECTS

Certified Research Specialist (CRS)[™]

- Visionary Research Association Inc. | Visionary Training Center

Final Examination Rating: 95.71%

- Core Competencies:
 - Research design and methodology
 - Ethical research practices
 - Statistical analysis and interpretation
 - Academic writing and publication preparation
- Demonstrates advanced competency in structured research production and scholarly communication

KEY RESEARCH & INNOVATION PROJECTS

- **EcoPET:** Microcontroller-based Pultrusion System for Recycling PET Bottles into 3D Printing Filament (Scopus-Indexed; Pitched & Matched at MentorMatch 2025).
- **FishDryPro:** Smart Fish Drying System (Most Voted Start-Up - ICT Summit 2025).
- **Advisory Projects:** Motorcycle Ignition via Facial Recognition with Telegram Notification; Brainwave-based Accident Prevention System; Egg Sorting with Breakage Detection.

RESEARCH ADVISING & MENTORSHIP EXPERIENCE

Research Adviser - Engineering and Technology Projects

Fellowship Baptist College | 2019 - Present

Served as research adviser for multiple undergraduate engineering and technology studies focusing on embedded systems, IoT applications, artificial intelligence, automation, and sustainable engineering solutions.

- **EGGSORT: A Multi-Microcontroller-Based System for Crack Detection, Weight Classification, and Packaging of Eggs with SMS Notification:** Microcontroller-based Pultrusion System for Recycling PET Bottles into 3D Printing Filament (Scopus-Indexed; Pitched & Matched at MentorMatch 2025).
- Focus: Embedded systems integration, automation, SMS-based monitoring, and quality control systems

AREA SITE SUPERVISOR (VISAYAS) / TECHNICAL SALES ENGINEER

2017 - 2019 Inter-Continental Systems Integrators Inc.

- **Regional Logistics:** Managed site equipment and inventory across multiple locations in the Visayas, ensuring 100% accountability for project materials.
- **Supply Chain Coordination:** Facilitated the procurement of specialized engineering components for large-scale technical installations.

CONTINUING PROFESSIONAL DEVELOPMENT & TRAINING

- AGM74 | Legacy of Excellence: Sustaining Innovation for Global Impact
- IECEP Annual General Meeting and Convention (20 CPD Points)
- Focus: Global Innovation Standards and Engineering Excellence.
- 74th IECEP Midyear Convention (2024) | Technical Sessions
- Specialization Track: Project Management & Academe
- Specialization Track: Research & Auxiliaries Track
- Specialization Track: Instrumentation, Control, & Power Electronics
- Advanced Engineering & Regulatory Seminars (IECEP NOC):
- Regulatory Compliance: Philippine Qualifications Framework (PQF) and Updates on Electronics Permit Implementation.
- Technical Design: Best Practices in Electronics Systems Design for Commercial Buildings.
- Safety & Systems: FDAS Cause and Effect Matrix and Emergency Control Functions.
- Emerging Tech: 5th Generation (5G) Technology System Design Overview.
- Solar Energy Photovoltaic Design (2017)
- Leadership Training: Student Council Alliance of the Philippines (2016).

- **EGGSORT: A Multi-Microcontroller-Based System for Crack Detection, Weight Classification, and Packaging of Eggs with SMS Notification:** Microcontroller-based Pultrusion System for Recycling PET Bottles into 3D Printing Filament (Scopus-Indexed; Pitched & Matched at MentorMatch 2025).
- Focus: Embedded systems integration, automation, SMS-based monitoring, and quality control systems
- **Drive ALERT (Driver Attention Level & Eye Response Tracker) Using B.R.A.I.N. Technology (Brainwave & Raspberry Pi AI Network)**
- AI-assisted driver monitoring, brainwave signal processing, and real-time safety systems
- **Design and Development of a Solar-Powered IoT-Based Arduino Egg Incubator System**
- Focus: Renewable energy integration, IoT monitoring, and agricultural automation systems
- **Arduino-Based CNC Styrofoam Cutting System**
- Focus: CNC automation, Arduino-based motion control, and fabrication systems

STATISTICAL & RESEARCH SUPPORT EXPERIENCE

IN-HOUSE STATISTICIAN / RESEARCH DATA ANALYST

Fellowship Baptist College | (2025–Present)

- Provided statistical analysis support for undergraduate and senior high school research studies
- Assisted in experimental design, data interpretation, and hypothesis testing
- Applied quantitative methods for research validation and result interpretation
- Supported thesis formatting, research methodology refinement, and data presentation

SENIOR HIGH SCHOOL RESEARCH ADVISING (STEM / APPLIED SCIENCE)

- **Repurposing Used Cooking Oil into Sustainable Lamp Fuels: A Comparative Study of Oil Types as Liquid Light Sources**
- Award: 3rd Best Research – Campus Colloquium
- Focus: Waste valorization, alternative fuels, and sustainability research
- **BINDUINO: A Sensor-Integrated Arduino-Based Smart Bin for Automated Solid Waste Segregation**
- Focus: Smart waste management, sensor integration, and environmental monitoring
- **Enhancing Solar Panel Photovoltaic Efficiency Using Mirror and Lens-Based Light Concentrators**
- **Best Research Paper (Campus/Institutional Colloquium)**
- Focus: Renewable energy optimization and solar efficiency enhancement
- **Growth, Morphological Traits, and Stress Responses of Lactuca sativa 'Green Ice' Under Varying Artificial Light Photoperiods**
- Focus: Controlled environment agriculture and plant physiology
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- **Innovative Use of Rice-Husk Waste for Manufacturing Eco-Friendly Food Packaging**
- Focus: Sustainable materials and agricultural waste valorization

PROFESSIONAL REFERENCES

Engr. Jennee Faith Joy Resano, ME

Dean, College of Engineering and Computer Studies,
Fellowship Baptist College
0928-367-9652

Engr. Japeth Peloton, ME

Engineering Dept., Inter-Continental Systems
Integrators Inc.
0917-244-5286

Engr. Ira Gayumali, MECE

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0918-963-6084

Rhea S. Calvo MSCS.

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